



CL 299 presentation 4

LNP Manufacturing and Process Simulation

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Key Process Parameters

Mixer Conditions

Parameter

Value

Flow rate ratio (aqueous : organic)

3 : 1

Total mixer flow

12 mL/min

Ethanol fraction after mixing

25 %

Dilution Step

Parameter

Value

Dilution buffer flow

36 mL/min

Total flow after dilution

48 mL/min

Ethanol concentration

6.25 %

TFF Operation

Parameter

Membrane MWCO

TMP

Crossflow rate

Concentration factor

Value

100 kDa

1 bar

30–100 mL/min

6×

Material Balance Key Streams

Feed Streams

Stream

Organic lipid solution

Aqueous mRNA solution

Dilution buffer

Flow

3 mL/min

9 mL/min

36 mL/min

Total Feed to TFF: 48 mL/min

Final Product Flow Rate: 7.5 mL/min

Composition:

- Lipids = 30 mg/min
- mRNA = 4.5 mg/min
- Buffer salts = 102 mg/min

Overall Balance:

Input = 48 mL/min

Output = 48 mL/min

The Nanoparticle Structure (Primary Lipids)

Ionizable Lipids

- **Definition:** Lipids that change their electrical charge based on the surrounding pH (acidic vs. neutral).
- **Use:** Bind and encapsulate the negatively charged RNA payload, and later help release it inside the target cell.

Helper Phospholipids

- **Definition:** Structural lipids that form the foundation of standard cell membranes.
- **Use:** Provide structural integrity to the LNP and help the nanoparticle fuse with the target cell's membrane.

Sterols

- **Definition:** Rigid, ring-structured lipid molecules (e.g., cholesterol).
- **Use:** Fill gaps between other lipids to regulate the structural fluidity and stability of the nanoparticle core.

PEG Lipids

- **Definition:** Lipids attached to polyethylene glycol (PEG) polymer chains.
- **Use:** Coat the LNP surface to prevent particles from clumping together and hide them from the immune system.

Manufacturing & Processing Agents

Organic Solvents

- **Definition:** Carbon-based liquids (e.g., ethanol, methanol) capable of dissolving hydrophobic materials.
- **Use:** Dissolve the lipids so they can be mixed with the water-soluble RNA during the initial nanoparticle formation process.

Buffer Acids & Salts

- **Definition:** Chemical systems designed to maintain a specific, constant pH in a solution.
- **Use:** Control the pH during mixing to ensure ionizable lipids carry the necessary positive charge to efficiently bind the RNA.

Storage & Protection Excipients

Cryoprotectants

- **Definition:** Specific sugars or polyols (e.g., sucrose, trehalose) that protect physical structures at freezing temperatures.
- **Use:** Prevent ice crystals from physically piercing and destroying the fragile lipid spheres during deep-freeze storage.

Antioxidants

- **Definition:** Compounds that inhibit oxidation reactions.
- **Use:** Protect lipids and RNA from oxidative degradation, extending the overall shelf life of the formulation.

Chelating Agents

- **Definition:** Molecules (e.g., EDTA) that bind tightly to free, reactive metal ions.
- **Use:** Trap trace metals that would otherwise catalyze drug degradation or activate enzymes that destroy RNA.

Final Formulation & Delivery Excipients

Surfactants

- **Definition:** Surface-active agents (e.g., Polysorbate 20) that lower the surface tension of liquids.
- **Use:** Prevent the nanoparticles from sticking to the walls of glass vials, syringes, and manufacturing tubing.

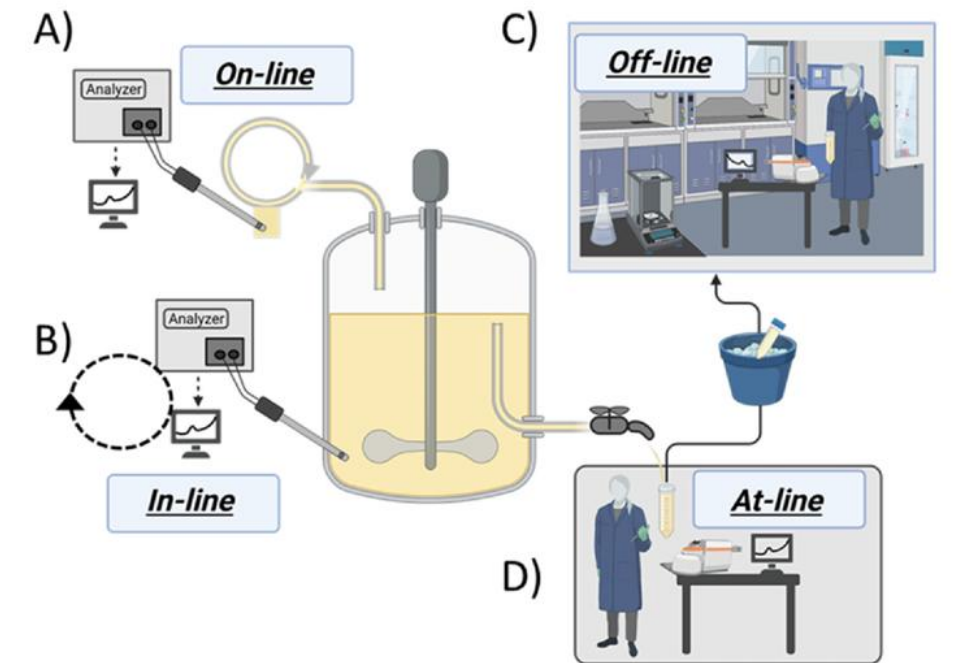
Salts

- **Definition:** Standard ionic compounds (e.g., Sodium Chloride).
- **Use:** Ensure the final liquid suspension matches the salt concentration of human blood to prevent cell damage and pain upon injection.

Stabilizers

- **Definition:** Viscosity-modifying or structure-supporting agents (e.g., Glycerol).
- **Use:** Keep the nanoparticle suspension evenly mixed and prevent the particles from settling or aggregating over time.

Level	Sample removal	Real-time	Control capability
Offline	Yes	No	Low
At-line	Yes	Limited	Medium
Online	Automated	Near real-time	High
Inline	No	Real-time	Highest



INLINE

- **Near-infrared Spectroscopy (NIR)**
this technique measure molecular vibrations → a fiber-optic NIR probe is installed directly inside the tablet coating equipment, measuring spectra continuously during coating.
- **Infrared Spectroscopy**
used to identify chemical bonding → For bioreactors.
- **Nuclear Magnetic resonance (NMR)**
used to determine molecular structures → flow water-proton NMR can be used as an inline PAT tool in continuous biomanufacturing to monitor protein concentration and aggregation in real time using water-proton relaxation signals.
- **Raman Spectroscopy**
analyse molecular structure and composition by measuring light scattering from molecules. → powder blending process
- **Dynamic Light Scattering (DLS)**
used to measure particle size.
- **Reflective Interference Spectroscopy**
measures interference patterns of reflected light to detect changes in surface thickness or refractive index and used as an optical biosensor to detect biomolecular binding by measuring changes in the interference of reflected light from a sensor surface

- **Surface Plasmon resonance (SPR)**

Surface plasmon resonance is an optical sensing technique that measures changes in refractive index at a metal surface to detect molecular interaction

- **Mass Spectroscopy**

provides information about the molecular weight → to monitor and characterize monoclonal antibodies in real time during purification.

- **NIR Turbidity**

uses Near-Infrared (NIR) light to determine how cloudy or particle-rich a liquid is.

- **IR Absorption**

Infrared (IR) absorption occurs when molecules absorb infrared radiation, causing changes in vibrational and rotational energy levels.

- **Zeta potential Analysis**

This technique determines the surface charge

- **PVM (Particle Vision and Measurement)**

captures real-time images of particles inside a process to monitor particle shape, size, and behavior.

Now in addition to all the techniques using inline classification, we have some tools that are used to measure some parameters-

1. pH sensors
2. Flow meters
3. Temperature probs
4. Conductivity sensors

ONLINE

- **NIR** → online NIR spectroscopy with an adaptive model-updating strategy to monitor industrial processes in real time as a PAT tool.
- **Infrared Spectroscopy & Mass Spectroscopy**
methanol electro-oxidation on a Pt catalyst using in-situ IR spectroscopy and online mass spectrometry to monitor reaction intermediates and products during the reaction.
- **NMR** → flow-NMR as an online PAT tool to monitor chemical reactions in real time by measuring reactant and product concentrations directly in the flowing reaction mixture.
- **Raman Spectroscopy**
- **IR absorption** → IR absorption at 5.85 μm to detect oxidation in lubricating oil and proposes a compact sensor for online oil condition monitoring.
- **High-Performance Liquid Chromatography (HPLC)**
On-line possible via auto-samplers for drug loading, encapsulation efficiency, or degradation monitoring.

- **Fourier Transform Infrared Spectroscopy (FTIR)**

online FT-IR spectroscopy as a PAT tool to monitor glucose consumption and ethanol production in yeast fermentation in real time.

- **Photometric stereo imaging**

Photometric stereo imaging reconstructs surface shape by analyzing images taken under different lighting angles.

ATLINE

- **Zeta potential analysis**
- **Fluorescence** → Atomic fluorescence spectrometry is a specific type of fluorescence technique used to detect trace metals by measuring fluorescence from atoms.
- **NIR** → During fermentation, samples were taken from the bioreactor and analyzed using Near-Infrared Spectroscopy (NIR) to estimate glucose and biomass using PLS models.
- **Liquid Chromatography** → at-line RP-UPLC (liquid chromatography) to monitor protein concentration during microfiltration.
- **Raman Spectroscopy** → portable Raman spectroscopy used as an at-line PAT tool to characterize carbon nanomaterials and monitor their structural quality during manufacturing.

OFFLINE

- **NMR , FTIR**
- **Fluorescence**
- **(Bio)chemical Analysers**
used to measure chemical or biological components
- **Confocal Spectroscopy (CICS)**
used for high-resolution nanoparticle characterization.
offline.
- **Consistent cryo-electron microscopy (Cryo-EM)**
This technique is used to visualize nanoparticle structure and morphology at very high resolution.
offline

- **Small-angle X-ray scattering (SAXS)**
This technique is used to analyze nanoscale structure and internal organization of particles using X-ray scattering.
offline
- **Optical Microscopy and Electron Microscopy**
helping to identify any issues like agglomeration or structural defects.
Offline.
- **Differential Scanning Calorimetry (DSC)**
(Thermal analysis, not real-time).
Offline.
- **X-ray Diffraction (XRD)**
This technique is used to determine crystal structure and phase composition by analyzing X-ray diffraction patterns.
Offline.
- **NIR & Raman Spectroscopy**
off-line NIR was used to correlate ribbon density with chemical absorption spectra, while **off-line Raman spectroscopy** was used to predict tablet hardness by analyzing surface smoothness and density.

Classification of techniques

Techniques	Offline	Atline	Online	Inline
Near-infrared Spectroscopy (NIR)	✓	✓	✓	✓
Infrared spectroscopy			✓	✓
Nuclear magnetic resonance (NMR)	✓		✓	✓
Raman spectroscopy	✓	✓	✓	✓
Dynamic light scattering (DLS)				✓
Fluorescence	✓	✓		
Surface plasmon resonance (SPR)				✓

Techniques	Offline	Atline	Online	Inline
Reflective Interference Spectroscopy				✓
Mass spectrometry (MS)			✓	✓
NIR Turbidity				✓
IR Absorption			✓	✓
Consistent cryo-electron microscopy (Cryo-EM)	✓			
Confocal Spectroscopy (CICS)	✓			
Small-angle X-ray scattering (SAXS).	✓			
Zeta potential Analysis		✓		✓

Techniques	Offline	Atline	Online	Inline
X-ray Diffraction (XRD)	✓			
High-Performance Liquid Chromatography (HPLC)			✓	
Fourier Transform Infrared Spectroscopy (FTIR)	✓		✓	
PVM (Particle Vision and Measurement)				✓
Photometric stereo imaging			✓	

Techniques	Offline	Atline	Online	Inline
Differential Scanning Calorimetry (DSC)	✓			
Liquid Chromatography		✓		
Optical Microscopy and Electron Microscopy	✓			
pH sensors				✓
Flow meters				✓
Temperature probs				✓
Conductivity sensors				✓

Resources

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THANK YOU!!!